

Late coefficients of three letter abelian squares

All fixed late orders and exact inverse Stokes sectors

A274600 research note

2 October 2026

Abstract

The coefficients in the inverse-power expansion of the three-letter abelian-square counts grow factorially. This note proves the explicitly labeled OEIS A274600 asymptotic conjecture and gives an effective expansion to every fixed late order. The proof uses the classical three-step random-walk density to identify the exact Borel transform, proves its required complex continuation, and extracts the logarithmic singularity by a keyhole contour. It also establishes the exact physical median and Stokes jump, Lambert-function index inversions, and a convergent generator of all fixed inverse exponential sectors. Exact-arithmetic replay and numerical checks accompany the proof. Novelty is stated only within the bounds of the documented source search.

1 Target, notation, and principal result

Write

$$M_N = \sum_{a+b+c=N} \binom{N}{a, b, c}^2 = \sum_{k=0}^N \binom{N}{k}^2 \binom{2k}{k} \quad (\text{A002893}).$$

Let a_n be the uniquely determined Poincaré coefficients in

$$M_N \sim \frac{C_0 9^N}{N} \sum_{n \geq 0} \frac{a_n}{(4N)^n}, \quad C_0 = \frac{3\sqrt{3}}{4\pi}. \quad (1)$$

These are A274600: $1, -1, 1, 2, 7, 59, 616, 6992, \dots$ The currently inspected OEIS entry, authored by Václav Kotešovec on 10 November 2016, explicitly conjectures

$$a_n \sim \frac{1}{\pi\sqrt{3}} \left(\frac{2}{\log 3} \right)^n (n-1)!. \quad (2)$$

This concerns the late coefficients of (1), not the leading asymptotic of M_N .

Put $L = \log 9$, $\kappa = 4/L = 2/\log 3$, $C = 1/(\pi\sqrt{3})$, and define

$$F(z) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right), \quad w(\delta) = \frac{(e^{\delta/2} - 1)^3(e^{\delta/2} + 3)}{16e^{\delta/2}},$$

$$G(\delta) = e^{3\delta/4}F(w(\delta)) = \sum_{j \geq 0} d_j \delta^j. \quad (3)$$

Theorem 1 (all fixed late orders). For each fixed positive integer r ,

$$a_n = C\kappa^n \Gamma(n) \left\{ \sum_{j=0}^{r-1} \frac{(-1)^j j! d_j L^j}{(n-1)_{\underline{j}}} + O_r(n^{-r}) \right\}. \quad (4)$$

Here $(n-1)_{\underline{j}} = (n-1)(n-2)\cdots(n-j)$, with the empty product equal to one. In particular, (2) is true. The first terms are

$$\begin{aligned} \frac{a_n}{C\kappa^n \Gamma(n)} &= 1 - \frac{3L}{4(n-1)} + \frac{9L^2}{16(n-1)(n-2)} - \frac{15L^3}{32(n-1)(n-2)(n-3)} \\ &\quad + \frac{135L^4}{256(n-1)\cdots(n-4)} - \frac{903L^5}{1024(n-1)\cdots(n-5)} \\ &\quad + \frac{63L^6}{32(n-1)\cdots(n-6)} + O(n^{-7}). \end{aligned} \quad (5)$$

The proof below also establishes the exact Borel transform, an exact median-Laplace representation, and a nonzero Stokes jump with action $L/4 = \log 3/2$ in the variable $1/(4N)$. Thus the exponentially small statement is not inferred merely from the divergent formal series.

2 The moment integral and the exact local Borel transform

For independent uniform θ, ϕ on $[0, 2\pi]$, put

$$T = |1 + e^{i\theta} + e^{i\phi}|^2 \in [0, 9].$$

Expanding the powers and integrating the Fourier monomials gives $M_N = \mathbb{E}[T^N]$. Multiplying the three unit vectors by a common independent phase does not change their resultant length. Consequently, if $p_3(x)$ is the radial density of the three-step planar uniform random walk,

$$M_N = \int_0^3 x^{2N} p_3(x) dx, \quad \rho(t) = \frac{p_3(\sqrt{t})}{2\sqrt{t}} \quad (6)$$

is the density of T .

The classical elliptic formula, in the modulus convention $K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-1/2} d\theta$, is

$$p_3(x) = \frac{\sqrt{x}}{\pi^2} \operatorname{Re} K\left(\sqrt{m(x)}\right), \quad m(x) = \frac{(x+1)^3(3-x)}{16x}. \quad (7)$$

For $1 < x < 3$, $m(x) \in (0, 1)$, so no real-part operation is needed. A primary reference is Borwein–Straub–Wan–Zudilin, *Densities of Short Uniform Random Walks*, Canadian Journal of Mathematics

64 (2012), 961–990, equation (1.2). Its section 3, equation (3.1), explicitly records the logarithmic singularity at $x = 1$.

For completeness, the part of (7) needed for the dominant endpoint can be obtained directly. The two-step resultant has density $2/(\pi\sqrt{4-r^2})$. Conditional on length r , adding a uniform unit vector makes its squared resultant arcsine-distributed on $[(r-1)^2, (r+1)^2]$. For $1 < x < 3$, substitution $s = r^2$ therefore gives

$$\rho(x^2) = \frac{1}{\pi^2} \int_{(x-1)^2}^4 \frac{ds}{\sqrt{s(4-s)((x+1)^2-s)(s-(x-1)^2)}} = \frac{K(\sqrt{m(x)})}{2\pi^2\sqrt{x}},$$

where the last equality is the standard four-real-root substitution for the complete elliptic integral. Multiplication by $2x$ gives (7) on that interval.

Set $t = 9e^{-u}$, $x = 3e^{-u/2}$. On $0 \leq u < L$, the normalized density becomes

$$f(u) = \frac{t\rho(t)}{C_0} = e^{-3u/4} F\left(m(3e^{-u/2})\right), \quad f(0) = 1. \quad (8)$$

For the real density on all $u \geq 0$, use $f_{\text{phys}}(u) = t\rho(t)/C_0$. Then

$$Q(N) := \frac{NM_N}{C_0 9^N} = N \int_0^\infty e^{-Nu} f_{\text{phys}}(u) du. \quad (9)$$

The part away from $u = 0$ is exponentially small for any fixed positive cutoff. Watson’s lemma applied to the analytic germ (8) yields

$$\boxed{a_n = 4^n n! [u^n] f(u)}. \quad (10)$$

Thus the order-one Borel transform of $\sum a_n z^n$, under the convention $\mathcal{B} \sum a_n z^n = \sum a_n \zeta^n / n!$, is exactly $f(4\zeta)$. Its nearest positive singularity will be $\zeta = L/4$.

3 Global continuation sufficient for coefficient transfer

This step is essential: a real endpoint expansion by itself does not prove (2).

The hypergeometric equation has possible singular values $m = 0, 1, \infty$. Its chosen solution $F(m)$ is regular at $m = 0$. The algebraic identities

$$m(x) = \frac{-x^3 + 6x + 8 + 3/x}{16}, \quad 1 - m(x) = \frac{(x-1)^3(x+3)}{16x} \quad (11)$$

give the following finite preimages under $x = 3e^{-u/2}$:

- $m = 1$: $u = L + 4\pi i k$, or $u = (4k + 2)\pi i$, $k \in \mathbb{Z}$
- $m = 0$: $u = 4\pi i k$, or $u = L + (4k + 2)\pi i$
- $m = \infty$: no finite u , because the exponential never vanishes

Fix any R with $L < R < 2\pi$. In $|u| < R$, the only singular-value preimages are $u = 0$, where the initial branch is analytic, and $u = L$. Hypergeometric analytic continuation and the monodromy theorem therefore continue (8) as a single-valued holomorphic function on

$$\{u : |u| < R\} \setminus [L, R). \quad (12)$$

To be precise about the removable initial point: the germ at zero is given by a convergent power series in $m(u)$; any sufficiently small loop around zero acts trivially on this germ. Continuation along paths avoiding zero and L , followed by the monodromy theorem in the slit disk, is therefore compatible with its original holomorphic extension through zero. Critical points of $m(u)$ that do not map to a singular value are merely pullback critical points, not singularities of the composite function.

In particular $u = L$ is the unique singularity on the circle of convergence and the function is analytic in a standard dented disk (a Δ -domain) around it. No assumption of Borel continuation beyond its justified domain has been made.

4 The logarithmic multiplier and coefficient transfer

The convergent zero-balanced hypergeometric expansion is

$$F(1-w) = \frac{1}{\pi} \sum_{k \geq 0} h_k w^k \left\{ 2\psi(k+1) - 2\psi(k + \tfrac{1}{2}) - \log w \right\}, \quad h_k = \frac{(1/2)_k^2}{(k!)^2}. \quad (13)$$

This is the $a = b = 1/2, m = 0$ case of DLMF 15.8.10. Put $\delta = L - u$. Since $x = e^{\delta/2}$, (11) yields

$$1 - m = w(\delta) = \frac{\delta^3}{32}(1 + O(\delta)). \quad (14)$$

The factor $w(\delta)/\delta^3$ is analytic and nonzero near zero. With the branch continued from $\delta > 0$,

$$\log w(\delta) = 3 \log \delta + \log(w(\delta)/\delta^3).$$

Combining (8), (13), and $e^{-3L/4} = 1/(3\sqrt{3})$, we obtain the exact local form

$$f(u) = -CG(L-u) \log(1-u/L) + H(u), \quad (15)$$

where H is holomorphic in a full neighborhood of L . The $\log L$ term has been absorbed in H . Equivalently, the analytic logarithmic multiplier is

$$g(u) = \frac{3}{\pi} e^{-3u/4} F(1 - m(3e^{-u/2})) = CG(L-u), \quad g(L) = C. \quad (16)$$

For fixed integer $j \geq 0$ and $n > j$, exact coefficient extraction gives

$$[u^n] \{ (1-u/L)^j \log(1-u/L) \} = \frac{(-1)^{j+1} j! L^{-n}}{n(n-1) \cdots (n-j)}. \quad (17)$$

Expand $G(L - u) = \sum d_j L^j (1 - u/L)^j$ and use (17). For a fully explicit remainder proof, choose R_0 with $L < R_0 < R$ sufficiently close to L that the convergent local form (15) applies along the cut from L to R_0 . A keyhole deformation of Cauchy's coefficient contour, using the jump $f_+ - f_- = 2\pi i C G(L - u)$, gives

$$[u^n]f(u) = C \int_0^{R_0-L} G(-v)(L+v)^{-n-1} dv + O(R_0^{-n}). \quad (17a)$$

The small circle around L vanishes because the singularity is logarithmic. On the outer contour the two boundary branches are bounded; their contributions are $O(R_0^{-n})$. Taylor's theorem gives $G(-v) = \sum_{j < r} (-1)^j d_j v^j + O_r(v^r)$. The remainder integral is bounded by the full beta integral, and for $n > j$,

$$\int_0^\infty v^j (L+v)^{-n-1} dv = L^{j-n} \frac{j! \Gamma(n-j)}{\Gamma(n+1)}. \quad (17b)$$

Extending each of the finitely many monomial integrals from $[0, R_0 - L]$ to $[0, \infty)$ changes it by an exponentially small quantity $O_j(R_0^{-n})$. Equations (17a)–(17b) prove

$$[u^n]f(u) = \frac{CL^{-n}}{n} \left\{ \sum_{j < r} \frac{(-1)^j j! d_j L^j}{(n-1)_{\underline{j}}} + O_r(n^{-r}) \right\}.$$

There is no hidden logarithmic loss in this bound. Multiplication by $4^n n!$ proves Theorem 1.

Expanding (3) gives

$$(d_0, \dots, d_8) = \left(1, \frac{3}{4}, \frac{9}{32}, \frac{5}{64}, \frac{45}{2048}, \frac{301}{40960}, \frac{7}{2560}, \frac{181}{172032}, \frac{24251}{58720256} \right). \quad (18)$$

These numbers are exact rationals. Equation (3) is an all-order, finite-at-each-order algorithm for all late correction coefficients.

5 An exact Stokes sector and exact median summation

Let $f_+(u)$ and $f_-(u)$, for real $u > L$, denote analytic continuation of the germ (8) past L in the upper and lower half u -planes, respectively, with a small semicircle and then continuation along the positive real axis. They are conjugates.

On the upper lip, $\arg \delta = -\pi$. The analytic continuation of $\log w$ therefore has imaginary part -3π , not $-\pi$. Equation (13) gives locally

$$f_+(u) - f_-(u) = 6ie^{-3u/4} F(1 - m(3e^{-u/2})) = 2\pi i g(u). \quad (19)$$

The right side continues without singularity along real $u > L$, since $m > 1$ and $1 - m < 0$. Both sides solve the same pulled-back second-order differential equation, so the local identity continues along the entire interval. More concretely, if $F(m + i0)$ denotes the principal hypergeometric upper boundary value, then near L , and hence on the entire interval by continuation,

$$f_+(u) = e^{-3u/4} \{F(m + i0) + 2iF(1 - m)\}, \quad f_-(u) = \overline{f_+(u)}. \quad (20)$$

Equivalently, standard hypergeometric continuation gives the especially transparent formulas

$$f_{\pm}(u) = e^{-3u/4} \{m^{-1/2}F(1/m) \pm 3iF(1-m)\}, \quad u > L. \quad (20a)$$

One way to verify the imaginary term is Pfaff's transformation $F(1-m) = m^{-1/2}F(1-1/m)$. The extra $2iF(1-m)$ in (20) is purely imaginary there. Equation (7) now proves the exact physical-median identity

$$f_{\text{phys}}(u) = \frac{f_+(u) + f_-(u)}{2} \quad (u > L). \quad (21)$$

It already holds trivially before L , where both continuations equal f .

As $u \rightarrow +\infty$, $x = 3e^{-u/2} \rightarrow 0$ and $m \asymp e^{u/2}$. The standard coincident-exponent hypergeometric estimate gives $F(m \pm i0), F(1-m) = O(m^{-1/2} \log m)$. Thus

$$f_{\pm}(u), g(u) = O((1+u)e^{-u}). \quad (22)$$

At $u = L$, the singularity is logarithmic and integrable. The lateral integrals

$$Q_{\pm}(N) = N \int_0^{\infty} e^{-Nu} f_{\pm}(u) du \quad (23)$$

are therefore absolutely convergent for real $N > 0$, with the obvious definition of $f_{\pm}$ before L . Equations (9) and (21) give

$$\boxed{Q(N) = \frac{Q_+(N) + Q_-(N)}{2}}. \quad (24)$$

These are actual lateral lip (equivalently, indented-contour) Borel–Laplace sums of (1), not only formal symbols: the Borel transform is (10), continuation on the lateral ray is established by (20), and growth is controlled by (22).

From (19), writing $u = L + v$,

$$\boxed{Q_+(N) - Q_-(N) = \frac{2i}{\sqrt{3}} 9^{-N} N \int_0^{\infty} e^{-Nv} G(-v) dv}. \quad (25)$$

Consequently, for every fixed integer $r \geq 1$, Watson's lemma gives the rigorous exponentially small expansion

$$Q_+(N) - Q_-(N) = \frac{2i}{\sqrt{3}} 9^{-N} \left\{ \sum_{j=0}^{r-1} \frac{(-1)^j j! d_j}{N^j} + O_r(N^{-r}) \right\}. \quad (26)$$

In particular,

$$Q_+ - Q_- \sim \frac{2i}{\sqrt{3}} 9^{-N} \left(1 - \frac{3}{4N} + \frac{9}{16N^2} - \frac{15}{32N^3} + \frac{135}{256N^4} - \frac{903}{1024N^5} + \cdots \right). \quad (27)$$

This proves an exact nonzero Stokes sector with multiplier $2i/\sqrt{3}$. In the small parameter $z = 1/(4N)$, the exponential is $e^{-A/z}$ with $A = L/4 = \log 3/2$. The same d_j control the late terms in (4) and the Stokes-sector fluctuations in (26), an explicit, rigorously derived resurgence relation.

Scope boundary. Equations (25)–(27) concern the difference of lateral sums and their exact physical median. They do not assert that the real moment function contains an independently adjustable real 9^{-N} term. Nor do they claim a complete global transseries, numerical Stokes constants at the farther complex singularities $u = \pm 2\pi i$, or uniform validity when the number of retained terms grows with N . Those questions require additional analysis.

6 Inverting the late-coefficient growth

Equation (4) implies $a_n > 0$ and $a_{n+1}/a_n \sim \kappa n$; the sequence is therefore strictly increasing eventually. The inverse statement here is first stated on exact values $y = a_n$, avoiding an unjustified choice of continuous interpolation.

Rewrite (4) as

$$a_n = C\kappa^n \Gamma(n) S(n), \quad S(n) \sim 1 + \frac{s_1}{n} + \frac{s_2}{n^2} + \cdots,$$

where, for example,

$$s_1 = -\frac{3L}{4}, \quad s_2 = -\frac{3L}{4} + \frac{9L^2}{16}. \quad (28)$$

Stirling's expansion yields

$$Y := \log \frac{y}{C\sqrt{2\pi}} = n \log \frac{\kappa n}{e} - \frac{1}{2} \log n + \sum_{j=1}^K \frac{c_j}{n^j} + O_K(n^{-K-1}), \quad (29)$$

with c_j computed by adding the formal logarithm of S to Stirling's logarithmic correction series. In particular,

$$c_1 = \frac{1}{12} - \frac{3L}{4}, \quad c_2 = -\frac{3L}{4} + \frac{9L^2}{32}. \quad (30)$$

Let $W = W_0$ be the positive real Lambert function and define

$$t_0 = \frac{Y}{W(\kappa Y/e)}, \quad h = \log(\kappa t_0), \quad b = \frac{\log t_0}{2h}. \quad (31)$$

Then, as $y = a_n \rightarrow \infty$,

$$\boxed{n = t_0 + b + \frac{b/2 - b^2/2 - c_1}{t_0 h} + O\left(\frac{1}{t_0^2 h}\right)}. \quad (32)$$

Indeed, $t_0 \log(\kappa t_0/e) = Y$. Substitute $t = t_0 + b + \varepsilon$ in (29): the leading residual is $(b^2/2 - b/2 + c_1)/t_0$, and the derivative is $h + O(t_0^{-1})$. The displayed ε cancels that residual; the remaining residual is

$O(t_0^{-2})$, proving (32) by the mean-value theorem. The derivative comparison uses an explicit smooth asymptotic model, not an assumed interpolation of a_n .

All fixed inverse orders. For arbitrary fixed K , let t_K be the unique sufficiently large solution of

$$t \log(\kappa t/e) - \frac{1}{2} \log t + \sum_{j=1}^K c_j t^{-j} = Y. \quad (33)$$

Its derivative is asymptotic to $\log t$, so the same mean-value argument gives

$$n = t_K + O_K \left(\frac{n^{-K-1}}{\log n} \right) \quad (y = a_n). \quad (34)$$

The coefficients are effectively generated by (3), falling-factorial expansion, logarithm, and Stirling's series. Newton iteration starting at (31) produces the corresponding explicit nested-Lambert expansion to any fixed order. For sufficiently large exact input a_n , rounding (32), or a sufficiently accurate t_K , recovers n .

7 Inverting the original moment sequence

There is a natural continuous interpolation here: $M(s) = \mathbb{E}[T^s]$, real $s > 0$. It is eventually strictly increasing, since the positive contribution from any interval $T \geq 1 + \varepsilon$ grows exponentially and dominates the bounded negative contribution to $M'(s)$ from $0 < T < 1$. Differentiated Watson estimates also show that the logarithmic derivative tends to L .

For large y , define

$$n_0 = -\frac{1}{L} W_{-1} \left(-\frac{C_0 L}{y} \right), \quad (35)$$

so that $C_0 e^{Ln_0}/n_0 = y$. Since

$$\log \frac{NM_N}{C_0 9^N} = -\frac{1}{4N} + \frac{1}{32N^2} + O(N^{-3}),$$

the actual inverse satisfies

$$\boxed{M^{-1}(y) = n_0 + \frac{1}{4Ln_0} + \left(\frac{1}{4L^2} - \frac{1}{32L} \right) \frac{1}{n_0^2} + O(n_0^{-3}).} \quad (36)$$

All fixed inverse orders follow by the same implicit-series/Newton procedure from (1), with error divided by a derivative asymptotic to L . This is an inversion of the actual positive moment function. Equations (35)–(36) do not by themselves claim a complete exponentially improved inverse transseries.

7.1 Exact inversion of the lateral sector family

The exponentially small structure can also be inverted, rather than only inverting its Poincaré part. Define

$$J(s) = s \int_0^\infty e^{-sv} G(-v) dv, \quad D(s) = \frac{3}{4\pi} \int_0^\infty e^{-sv} G(-v) dv = \frac{3J(s)}{4\pi s}, \quad \mathcal{M}_t(s) = M(s) + tD(s). \quad (36a)$$

The integral defines a holomorphic function for $\operatorname{Re} s > -1$, as does the moment integral M . Equations (24)–(25) imply exactly

$$\mathcal{M}_i(s) = \frac{C_0 9^s}{s} Q_+(s), \quad \mathcal{M}_{-i}(s) = \frac{C_0 9^s}{s} Q_-(s). \quad (36b)$$

Let $\phi = M^{-1}$, let $s = \phi(y)$ for sufficiently large positive y , and choose the inverse branch $n(t; y)$ of $\mathcal{M}_t$ that satisfies $n(0; y) = s$. This branch exists and is analytic in t on a disk of radius at least csy , where $c > 0$ is independent of large s . Indeed, fix $0 < r < 2\pi/L$. Uniform endpoint Laplace estimates give

$$\frac{M(s+w)}{M(s)} \longrightarrow e^{Lw} \quad \text{uniformly for } |w| \leq r, \quad D(s+w) = O(1/s).$$

On $|w| = r$, $e^{Lw} - 1$ has positive minimum modulus, and inside the circle it has exactly one zero, counted with multiplicity. Rouché's theorem first shows the same for $M(s+w) - y$, and then, for sufficiently small fixed c , for $M(s+w) + tD(s+w) - y$ uniformly on $|t| < csy$. The unique zero is simple, so the analytic implicit function theorem gives the claimed analytic branch, with the uniform bound $|n(t; y) - s| < r$. This also proves the bounds needed for the Cauchy remainder estimate below.

Lagrange–Bürmann inversion gives the exact convergent sector generator

$$\boxed{n(t; y) = \phi(y) + \sum_{k=1}^{\infty} \frac{(-t)^k}{k!} \left(\frac{d}{dy} \right)^{k-1} [\phi'(y) D(\phi(y))^k]}. \quad (36c)$$

For example, apply ordinary reversion to $z + tD(\phi(z)) = y$, then the generalized Lagrange formula to $\phi(z)$. Its germ at $t = 0$ is (36c), and analytic continuation on the established disk proves equality throughout the disk of convergence. For each fixed K , truncation at $k = K$, evaluated at either $t = i$ or $t = -i$, has rigorous error

$$O_K((sy)^{-K-1}). \quad (36d)$$

This follows directly from Cauchy's estimate on a fixed fraction of the disk $|t| < csy$; no growing-order asymptotic assertion is being used.

Each fixed k -sector has an all-fixed-order expansion in s^{-1} , effectively obtained from

$$\phi'(y) = \frac{1}{M'(s)}, \quad \frac{d}{dy} = \frac{1}{M'(s)} \frac{d}{ds}, \quad J(s) \sim \sum_{j \geq 0} (-1)^j j! d_j s^{-j}$$

and (1). The required differentiated remainder estimates follow by differentiating the absolutely convergent Laplace integrals, or by Cauchy estimates uniform on fixed disks centered at large positive s , retaining sufficiently many terms before taking a fixed number of derivatives. Its leading term is

$$\frac{(-t)^k}{k!} \partial_y^{k-1} [\phi'(y) D(\phi(y))^k] = -\frac{t^k}{kL} \left(\frac{3}{4\pi sy} \right)^k (1 + O_k(s^{-1})). \quad (36e)$$

Since $sy = C_0 9^s Q(s)$, these are genuine 9^{-ks} sectors. Thus (36c)–(36e), together with the all-order fluctuation algorithm, constitute a rigorous inversion of the exact lateral transseries family.

Writing $n_{\pm}(y) = n(\pm i; y)$, the first terms are

$$n_{\pm}(y) = s \mp i \frac{D(s)}{M'(s)} + O((sy)^{-2}), \quad n_+(y) - n_-(y) = -2i \frac{D(s)}{M'(s)} + O((sy)^{-3}). \quad (36f)$$

Odd/even cancellation in the second statement is justified by the convergent analytic t -series, not merely by informal conjugation. The first inverse Stokes sector therefore has the all-fixed-order expansion beginning

$$n_+(y) - n_-(y) = -\frac{3i}{2\pi Lsy} \left[1 + \frac{L^{-1} - 3/4}{s} + \frac{L^{-2} - L^{-1} + 9/16}{s^2} + O(s^{-3}) \right] + O((sy)^{-3}). \quad (36g)$$

The physical inverse is $s = \phi(y)$. Inversion does not commute with taking the median: from the even terms of (36c),

$$\frac{n_+(y) + n_-(y)}{2} = s - \frac{1}{2} \frac{d}{dy} \left[\phi'(y) D(\phi(y))^2 \right] + O((sy)^{-4}), \quad (36h)$$

whose leading correction is $9/(32\pi^2 L s^2 y^2)$. This explicitly distinguishes the physical inverse from the mean of the two lateral inverses. The exact sector generator concerns this specified lateral family; it does not assert that all possible distant-sheet sectors of the original Borel transform have been classified.

Editorial note (ProveIt, 2026-10-02). The three inversions (31)–(34), (35)–(36) and (36c) are cases of the inversion apparatus of ProveIt’s canonical transseries volume, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion* (Part X, chapter `p0:sec:top`), which the article does not cite (its repository searches in Section 9 looked for the sequence numbers and “honeycomb”); the method is the volume’s, and only the coefficients are new to the repository. The volume’s source is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`.

Late coefficients. The seed (31) solves the volume’s factorial core, Proposition J.27 (`p0:prop:factorial-core`), with $\kappa = 1$, $d = \log \kappa - 1$. In the displacement $n = t_0(1+E)$, (33) becomes $hE + [(1+E) \log(1+E) - E] + f(t, E) = 0$ with $t = 1/t_0$ and $f = -\frac{1}{2}t(\log t_0 + \log(1+E)) + \sum_{j \geq 1} c_j t^{j+1} (1+E)^{-j}$, which is its reversion about an arbitrary core, Theorem J.33 (`p0:thm:core-reversion`), with $\Lambda = h$; its first two coefficients are b and $(b/2 - b^2/2 - c_1)/h$, the terms of (32), which the editors recomputed symbolically. Recovering n by rounding when $y = a_n$ is part (3) of the volume’s staircase theorem, Theorem J.43 (`p0:thm:staircase`): an approximation within $\eta < \frac{1}{2}$ of the interpolated inverse at $y = A_n$ rounds to n .

Original moments. Since $\log M(s) = \log C_0 + Ls - \log s + Q(1/s)$ with $Q(t) = -t/4 + t^2/32 + O(t^3)$, (35) is the volume’s exact solution of the dominant block, Theorem J.23 (`p0:thm:lambert-core`), with $a = L$, $b = -1$, the large root lying on the branch W_{-1} by its branch rule (Corollary J.25, `p0:cor:branch-rule`); and (36) is its all-orders reversion around the Lambert core, Theorem J.29 (`p0:thm:lambert-centered`), whose Bell recurrence gives $c_1 = -q_1/a = 1/(4L)$ and $c_2 = -(bc_1 + q_2)/a = 1/(4L^2) - 1/(32L)$, the coefficients of (36).

Sectors. (36c) is the volume's perturbed inversion, Theorem J.21 (`p0:thm:perturbed-inversion`), with $F = M$, $E = D$, $\varepsilon = t$ and $\mu_0 = \phi(y)$: its derivation $\mathcal{D} = F'(\mu)^{-1} d/d\mu$ is d/dy , so its term of order k (index M there), $\frac{(-t)^k}{k!} \mathcal{D}^{k-1}(D(\mu_0)^k/M'(\mu_0))$, is the k th term of (36c). The theorem is local; the radius csy of the t -disk, and with it the remainder (36d), is the article's own Rouché estimate.

8 Exact replay and independent finite checks

The known recurrence for M_N is

$$(N+1)^2 M_{N+1} - (10N^2 + 10N + 3)M_N + 9N^2 M_{N-1} = 0. \quad (37)$$

Substitute the formal series (1), put $B(z) = \sum_{n \geq 0} a_n z^n / 4^n$, and compare coefficients in

$$9(1+z)B\left(\frac{z}{1+z}\right) - (10+10z+3z^2)B(z) + \frac{1}{1-z}B\left(\frac{z}{1-z}\right) = 0. \quad (38)$$

This gives the exact triangular recurrence

$$8na_n = \sum_{k=0}^{n-1} 4^{n-k} \left\{ 9 \binom{1-k}{n+1-k} + \binom{n+1}{k} \right\} a_k - 12a_{n-1}, \quad n \geq 1, \quad a_0 = 1. \quad (39)$$

The binomial coefficient with negative top is the ordinary generalized integer binomial. Uniqueness of the asymptotic coefficients and recurrence matching justify (39). This is not used as a substitute for the analytic proof of late growth.

`verify.py` uses Python integer/fraction arithmetic to:

1. match all 22 displayed source terms exactly
2. compute the germ (8) independently of (39), matching 31 coefficients exactly
3. generate a_n through $n = 500$, checking integrality at every recurrence step
4. compute $d_0, \dots, d_{15}$ exactly
5. compare the late approximations against exact a_n at $n = 20, 40, 80, 160, 320, 500$

The replay output is `verification.json`; the generated integers are `coefficients_0_500.txt`. For example the exact-to-leading ratio at $n = 500$ is approximately $0.996708446210591670313689669302$; retaining eight falling-factorial terms leaves normalized error approximately 2.45029×10^{-18} . These finite numerical checks support implementation correctness; the asymptotic theorems follow from sections 2–5.

n	$a_n/(C\kappa^n\Gamma(n))$	Error after 4 terms	Error after 8 terms
20	0.920460390900	$1.08 \cdot 10^{-4}$	$1.79 \cdot 10^{-6}$
40	0.959493066036	$5.65 \cdot 10^{-6}$	$2.92 \cdot 10^{-9}$
80	0.979570825175	$3.25 \cdot 10^{-7}$	$7.75 \cdot 10^{-12}$
160	0.989742589835	$1.95 \cdot 10^{-8}$	$2.52 \cdot 10^{-14}$
320	0.994860728387	$1.20 \cdot 10^{-9}$	$8.99 \cdot 10^{-17}$
500	0.996708446211	$1.99 \cdot 10^{-10}$	$2.45 \cdot 10^{-18}$

The errors in this table are differences of normalized quantities, not relative errors of the original moments. Each column keeps a fixed number of terms.

The second replay, `verify_analytic.py`, evaluates the density integral using a complementary-AGM form that preserves the cubic zero at $x = 1$. At 70-digit working precision its relative errors against the exact moments at $N = 1, 2, 5, 10, 20$ are below 10^{-50} . These are floating-point consistency checks, not interval-certified error bounds. Formula (32) estimates the late index $n = 500$ as $500.0000002664748\dots$; formula (36) estimates the original moment index $N = 30$ as $30.0000005839233\dots$. Full numerical values are in `analytic_verification.json`.

9 Provenance and scope of the novelty check

Primary sources used:

- [OEIS A274600](#): target sequence, scaling, displayed terms, and explicit conjecture
- [OEIS A002893](#): moment/count identity, honeycomb-return interpretation, recurrence, and older leading asymptotic
- [Borwein–Straub–Wan–Zudilin \(2012\)](#): equations (1.2), (3.1), and (3.2); elliptic density, its logarithmic singularity, and its relation to the same moments
- [NIST DLMF 15.8.10](#): convergent zero-balanced hypergeometric logarithmic expansion
- [NIST DLMF 15.12](#): hypergeometric large-variable behavior
- [Richmond–Shallit, Counting Abelian Squares](#): the original abelian-square leading asymptotic is established background
- [Burnette–Wong, Abelian Squares and Their Progenies \(2016\)](#): broader existing asymptotic context
- [Bašić–Fowler–Pickup–Potočnik \(2026\), Random walks and the electronic structure of graphene](#): checked as a recent A002893 reference; its inverse-bandgap expansion uses the original moments, not the late coefficients in A274600
- [The Stack Exchange discussion linked by A274600](#): 2016 discussion of the leading sum asymptotic; not a located proof of the late-coefficient assertion

Exact-ID searches for A274600 with proof, asymptotic, factorial, ProveIt, and OEIS Open did not locate a proof. Searches for A002893 and honeycomb/three-step late asymptotic coefficients did locate the classical density literature and general resurgence literature, but no source proving this exact late coefficient formula. The evidence supports “a proof of the explicitly labeled OEIS conjecture, with an all-fixed-order and exact Stokes-jump strengthening”; it does not establish that the mathematical mechanism was previously unknown or rule out unpublished or differently indexed results. The elliptic density, its van Hove logarithm, and Watson/singularity-transfer methods are classical.

The source b-file was linked by the entry but inaccessible through the checked HTTP/web paths; the finite source agreement is therefore explicitly limited to the 22 displayed terms, not all 81 b-file entries. The ProveIt default-branch code searches for A274600, A002893, and honeycomb each returned zero results. The inspected complete Analysis, Combinatorics, and Oeis subtree listings (5,850, 462, and 9 entries respectively; each marked untruncated) contain no exact target-ID path. Their parent root tree was 4b874cea0012c51a6841ad9c58f6e20fa57c71da. This is a bounded duplicate check, not a full-text search of every file at every historical revision; details are in SOURCES.md. No external submission or repository change has been made.

10 Further questions

The argument settles the labeled leading conjecture and gives all fixed late orders, the exact positive-axis Stokes sector, and its inverse-sector family. Several extensions remain outside its scope.

1. Continue the selected Borel branch to the next singularities at $u = \pm 2\pi i$, determine their sheet-dependent connection constants, and isolate their effects on late coefficients after treating the dominant cut
2. Establish optimal truncation, quantitative exponentially improved remainder estimates, and uniform Stokes smoothing when the truncation order grows with the large parameter
3. Investigate all-index arithmetic properties of the normalized coefficients; the present replay checks integrality through index 500 but does not supply a general integrality proof
4. Compare the same density-to-late-coefficient mechanism for larger alphabets, where different critical values and singularity types may govern the factorial scales

These are research directions, not assertions that the corresponding questions are new or unresolved in all existing literature.